

Web Analytics & Information Retrieval

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What is Web analytics?

- Web Analytics is the process of **collecting, analyzing, and reporting** web data to **understand and optimize web usage** by **tracking, reviewing and reporting the data** generated by their use of the site and its components, such as its webpages, images and videos.
- It helps in tracking and understanding user behavior on websites.

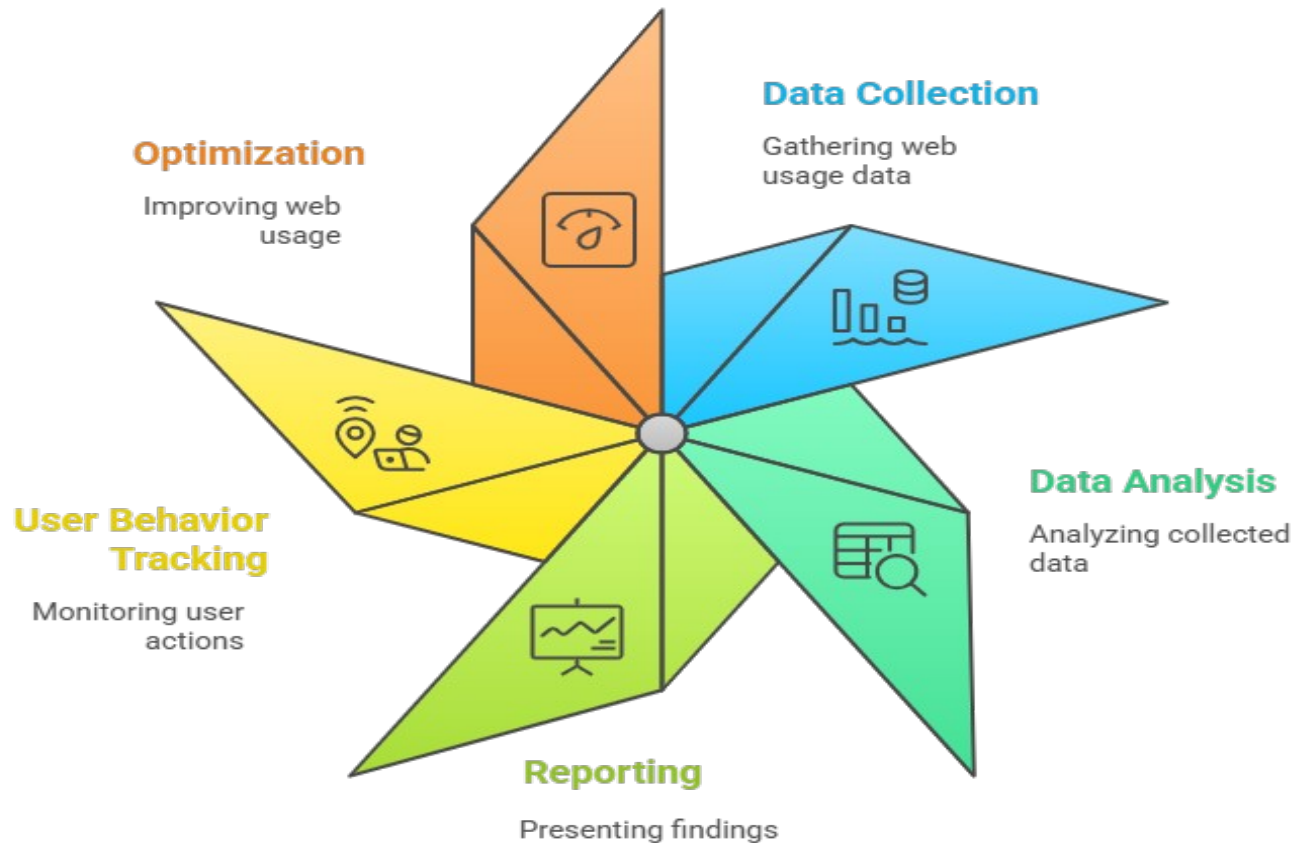
Web Analytics Goal

The goal is to provide actionable insights that can help the site owner **optimize** the website for users and **meet business objectives** such as increased traffic, conversions or sales.

Process

- Collecting user behavioral data on a website or web application and analyzing that data for a specific purpose facilitates making decisions to achieve that purpose.
- Analytics **tools communicate with web browsers** to track and analyze online behavior,
- Then **generate results in human-readable formats** like tables, charts or graphs.

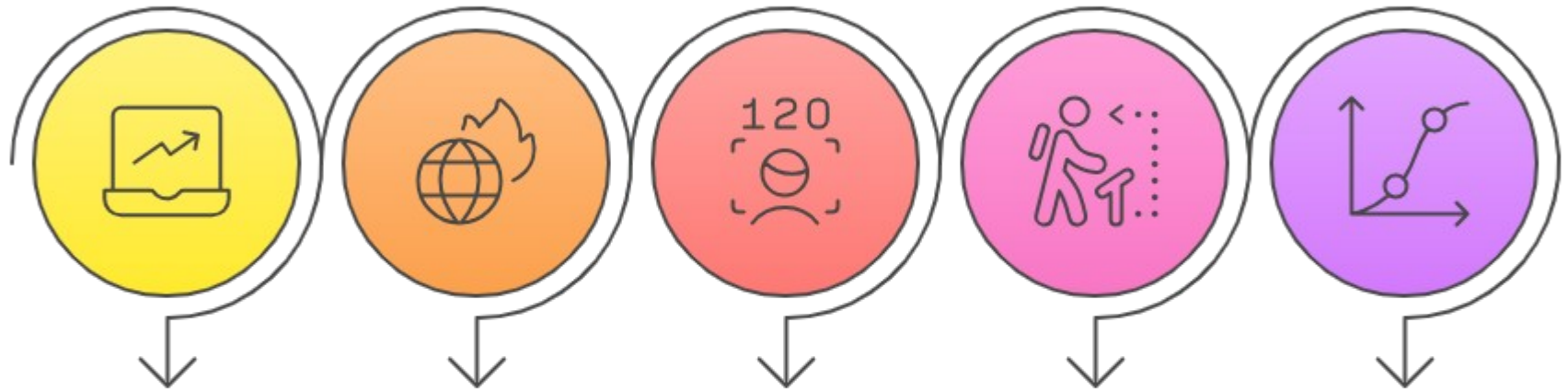
Understanding Web Analytics



Data collected by the tools

Data the tool collects may include traffic sources, referring sites, page visits or views, paths taken and [conversion](#) rates,

Data collected by the tools



Traffic sources

Information about where traffic originates.

Referring sites

Websites that send visitors to the site.

Page visits

Number of times specific pages are visited.

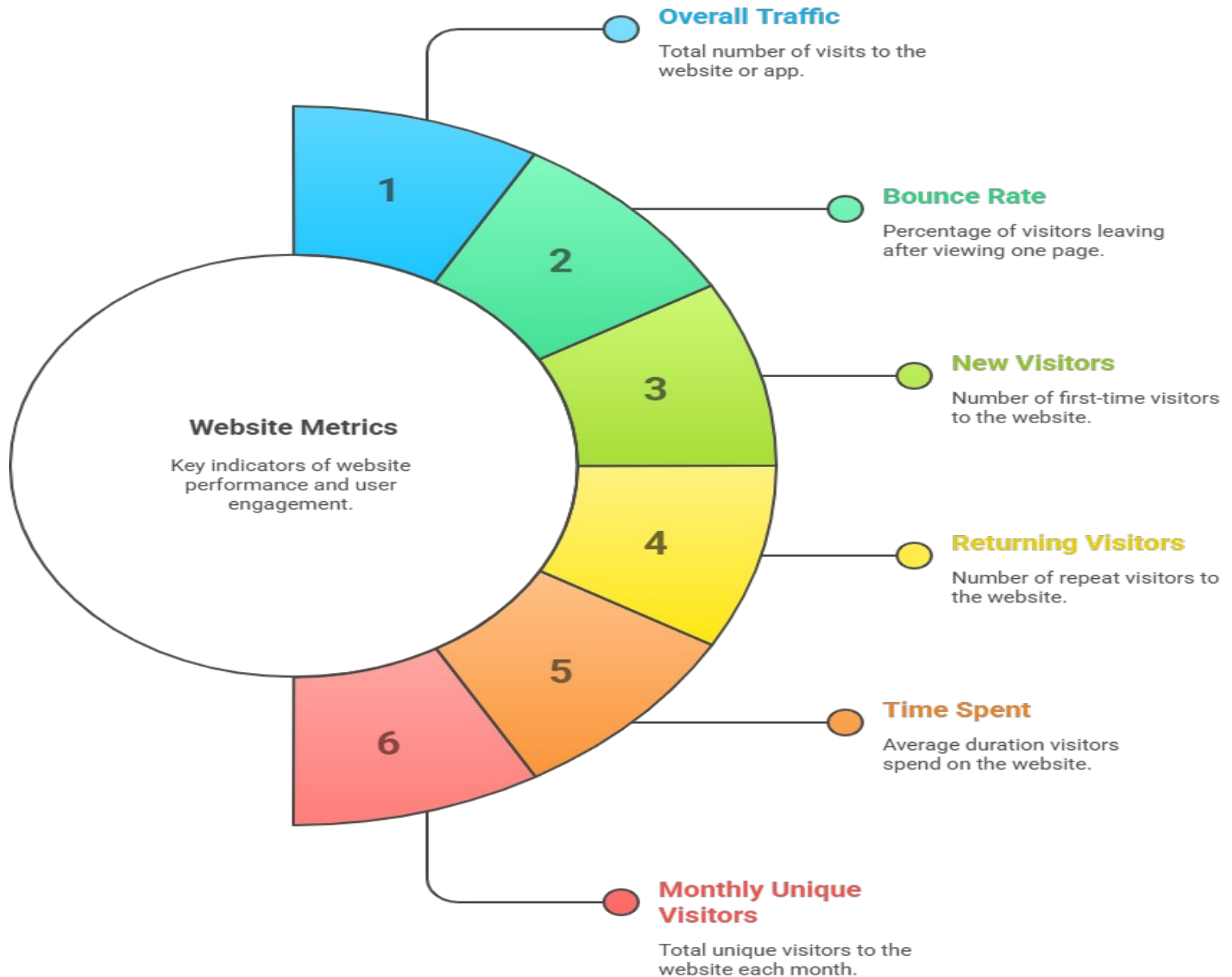
Paths taken

The sequence of pages a user visits.

Conversion rates

Percentage of visitors completing desired action.

Exploring Key Website Metrics



What is a metric?

- A metric is a category of quantifiable data relevant to a company's performance.
- Everything you track in your client's business is a metric.
- Metrics can be attached to almost every part of a client's standard business processes.
- Metrics help provide context for the performance of key business goals but are not critical to their success like KPIs are.

What are Website KPIs?

- Website KPIs ([key performance indicators](#)) **measure** your website performance and tie your marketing efforts to the overall business goals.
- Key performance indicators, or KPIs, **measure performance or progress based on specific business goals and objectives**. A pivotal element to consider is the word “key,” meaning they only track what is truly relevant to the company’s strategic decisions.
- A good KPI should **help you and your team understand if you are making the right decisions**. They **act as a map of business outcomes** and are the strategic indicators that will move the company forward.
- Example: An ecommerce website that wants to increase rankings in the top 10 search engine results (website KPI) might measure metrics like [website traffic](#), click rates in emails, and number of leads in the sales funnel.

KPI vs Metric

Key Performance Indicator (KPI)	Metric
Focused	Limited to one result or dimension
Bigger-picture	Granular
Represent key business goals	Measure specific tactics
Usually long-term goals	Usually shorter-term goals
Can be measured in various ways	Usually tied to specific data
Results-oriented	Focuses on specific business processes and problems
Often easier to understand	Needs interpretation

KPIs and metrics are often considered synonyms. But this is not how it actually works. While they are both quantitative measurements, **they are used for different purposes.** Simply put, KPIs need to be **exclusively linked to targets** or goals to exist, and **metrics just measure the performance of specific business actions** or processes.

Compare KPIs and Metrics for Strategic Insights



Bigger-picture
focus



Granular detail
focus



Long-term goals



Short-term
tactics



Various
measurement
methods



Specific data
ties



Results-oriented



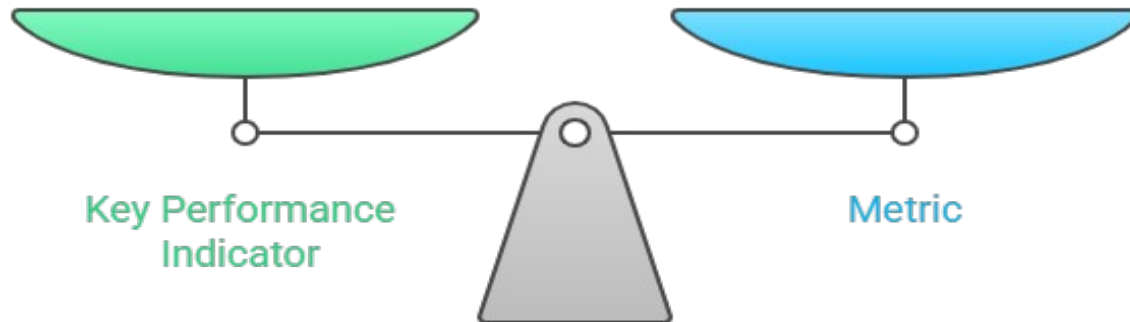
Process-focused



Easier to
understand

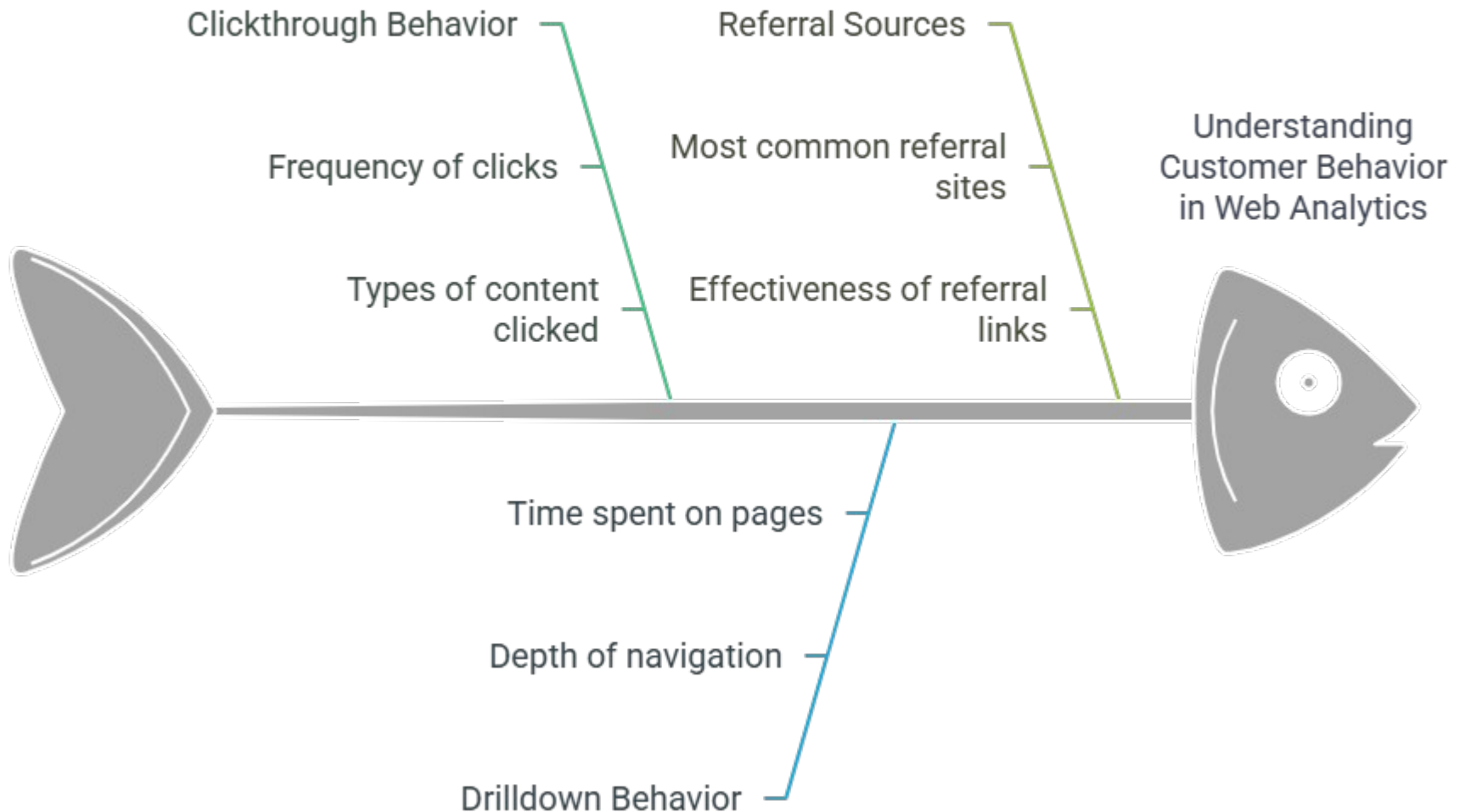


Needs
interpretation



- Web analytics may track customer clickthrough and drilldown behavior and determine the sites from which customers most often arrive.

Analyzing Customer Behavior in Web Analytics

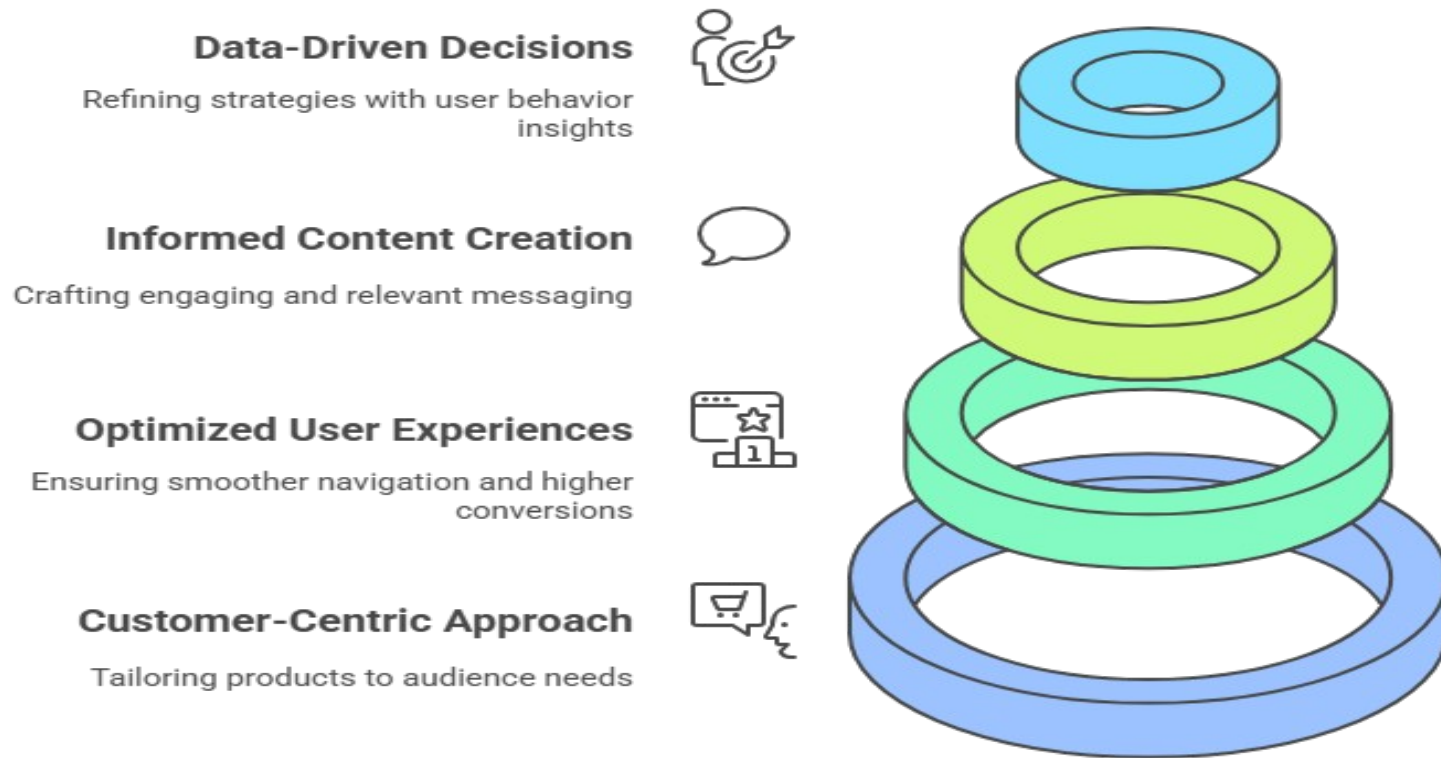


User Behaviour

- User behavior impact is the digital footprint left behind by your audience, and it is a critical compass for businesses to decipher.
- Every click, scroll, hover, and interaction carries a story, revealing the preferences, intent, and pain points of your visitors. In this era where personalized experiences are paramount, grasping user behavior is akin to holding the keys to customer satisfaction and loyalty.

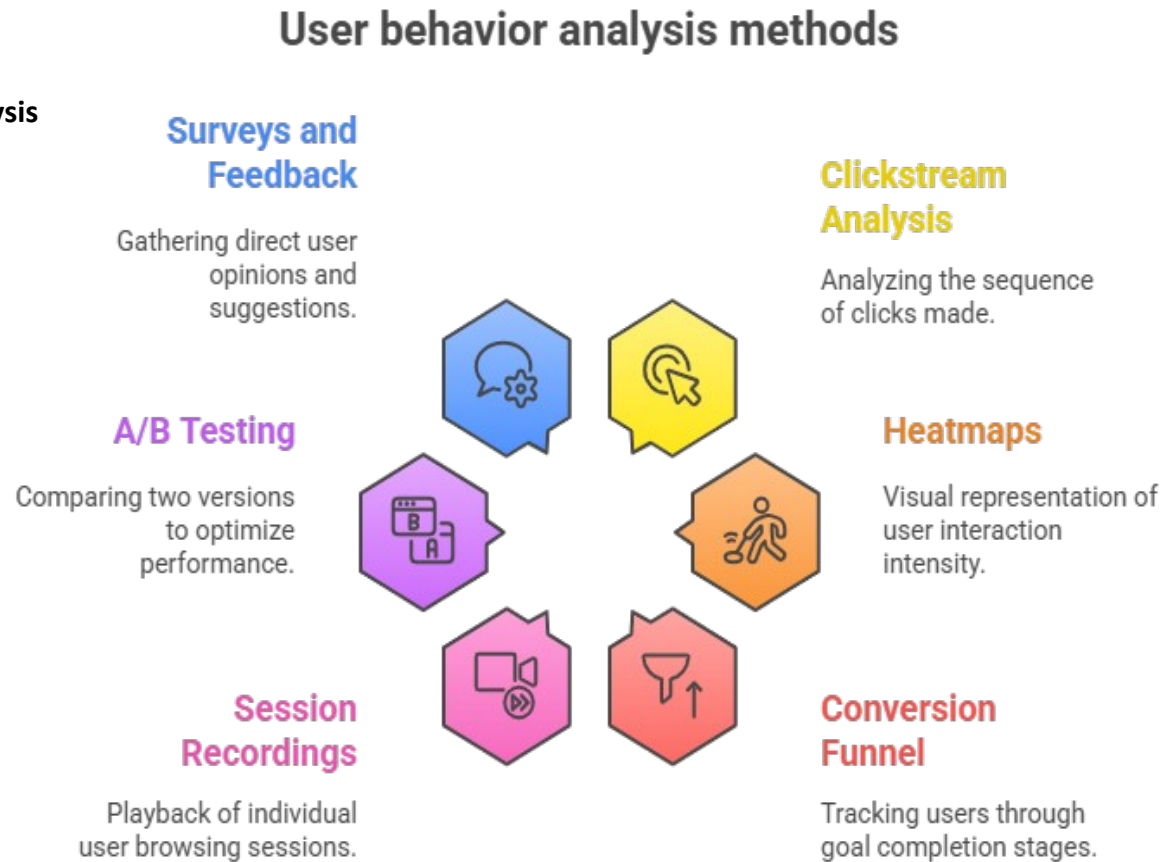
Why user behaviour is crucial?

User Behavior Insights Pyramid



Understanding/ Analyzing User Behaviour

- Clickstream Analysis
- Heatmaps
- Conversion Funnel Analysis
- Session Recordings
- A/B Testing
- Surveys and Feedback



Web Data Sources

1. **Website Traffic Data** - This is the primary source, providing information about website visits, user actions, and website interactions.
 - **Page Views:** The number of times a specific page was viewed.
 - **Click-Through Rates:** The percentage of users who clicked on a specific element.
 - **Time on Page:** The amount of time a user spent on a specific page.
 - **Bounce Rate:** The percentage of visitors who leave the website after viewing only one page.
 - **Traffic Sources:** Where visitors came from, such as organic search, social media, or direct traffic.

2. **Server Logs** - Web servers maintain detailed records of all requests and responses, providing a comprehensive view of website activity.
 - **IP Addresses:** The unique identifier of a visitor's computer.
 - **HTTP Requests:** Details about the data being requested by the user.
 - **Server Responses:** Details about the data sent back to the user by the server.

3. Tracking Codes - These small pieces of code are integrated into websites to collect user data.

- **Cookies:** Small files stored on a user's browser to track browsing behavior.
- **Tracking Pixels:** Invisible images embedded in web pages that track user visits.

4. External Data Sources - These can be integrated with website data to provide a more complete picture of user behavior.

- **Social Media Analytics:** Information about website traffic from social media platforms.
- **Advertising Platforms:** Data about campaign performance and user interactions.
- **Customer Feedback:** Qualitative data from surveys and feedback forms

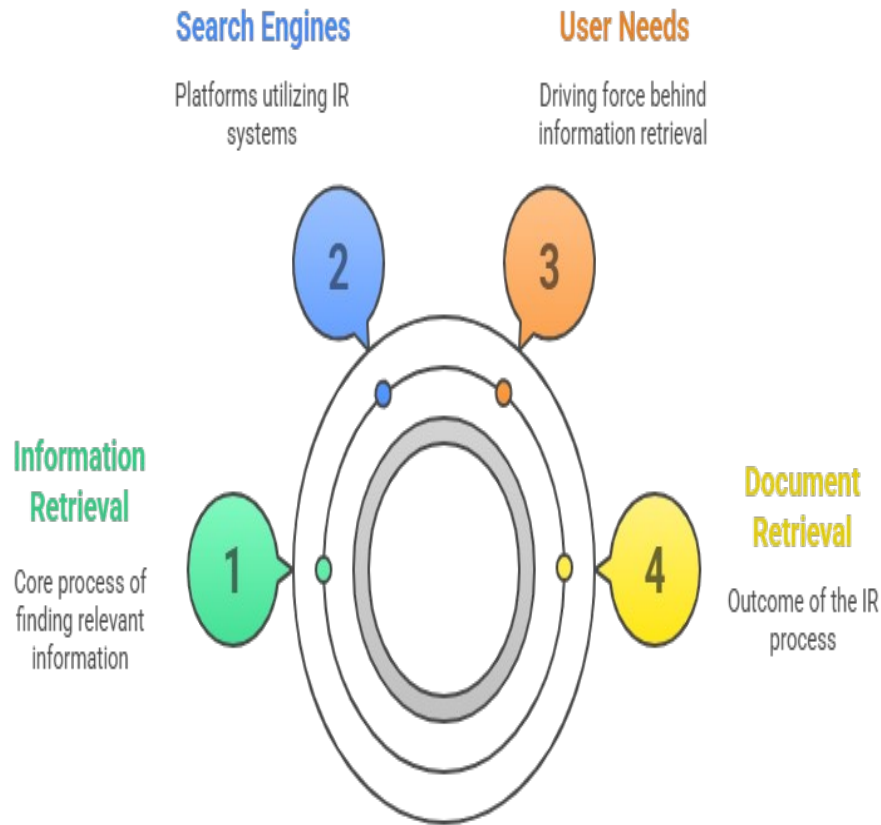
5. Internally Generated Data - This data is generated by the website itself.

- **Internal Logs:** Records of user interactions within the website.
- **Application-Level Data:** Data collected by application-level programs, such as JavaScript or PHP.

What is Information Retrieval?

- Information Retrieval (IR) helps to find relevant information from large collections of documents. It can be defined as **a software program that deals with the organization, storage, retrieval and evaluation of information from documents.**

Information Retrieval System



- Information Retrieval is a **core component of search engines** like Google or Bing.
- It focuses on helping users **find documents, files, web pages, or any form of information** that matches their needs, based on keywords or phrases they provide.
- IR systems **do not return exact answers** like databases do—they **rank and retrieve documents** that are **most relevant**.

Basics of Information Retrieval

1. Indexing

Indexing is the process of organizing and structuring documents or web content so they can be searched quickly and efficiently.

How it works:

- Similar to the index in a book.
- A search engine creates an **inverted index**: a mapping of keywords to the list of documents they appear in.

Purpose:

- Enables fast retrieval.
- Reduces the time required to find relevant documents.

2. Querying:

Querying is the process of submitting a search term or phrase to an IR system to find relevant documents.

Types of Queries:

- **Keyword Query** – E.g., web analytics tools
- **Boolean Query** – E.g., analytics AND marketing
- **Phrase Query** – E.g., "information retrieval"
- **Wildcard Query** – E.g., retriev* to match "retrieve", "retrieval", etc.

Purpose:

- Translates user needs into a form the system can understand.
- Retrieves documents that contain relevant content.

3. Ranking

Ranking is the process of ordering the search results based on their relevance to the user's query.

How it works:

- IR systems score each document based on a **relevance function**.
- Higher scores = more relevant = shown at the top.

Common Ranking Algorithms:

- **TF-IDF (Term Frequency–Inverse Document Frequency):**
Gives importance to words that are frequent in a document but rare across all documents.
- **BM25:**
A probabilistic ranking model, refined version of TF-IDF.
- **PageRank** (used by Google):
Ranks pages based on the number and quality of links pointing to them.

Example:

A search for "web analytics" might return 100 documents. The ranking algorithm will sort and display the most relevant top 10 on the first page.